Bright Exercise

SELECTING & SORTING

Table: employees

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Q1: Retrieve all columns from the employees table

```
SELECT *
FROM employees;
```

Output:

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Q2: Find all unique departments

```
SELECT DISTINCT department
FROM employees;
```

Output:

department
IT
HR
Finance
Marketing

Q3: Retrieve first and last names ordered by salary descending

```
SELECT first_name, last_name
FROM employees
ORDER BY salary DESC;
```

Output:

first_name	last_name
Eva	Maidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

Q4: Retrieve the top 3 highest-paid employees

```
SELECT id, first_name, last_name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

Output:

id	first_name	last_name	salary
5	Eva	Maidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

FILTERING with WHERE

Q5: Find employees in the IT department

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Maidoo	IT

```
SELECT id,
       first_name,
       last_name,
       department
FROM employees
WHERE department = 'IT';
```

Q6: Find employees in Finance with salary > 60000

```
SELECT id, first_name, last_name, department, salary
FROM employees
WHERE department = 'Finance'
AND salary > 60000;
```

Output:

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65000

Q7: Find employees in HR or Marketing

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('HR', 'Marketing');
```

Output:

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

Q8: Find employees not in IT

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department != 'IT';
```

Output:

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	marketing
3	Cathy	Zulu	Finance

Q9: Find employees in IT, HR or Finance using IN

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

Output:

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Maidoo	IT

Q10: Find employees in IT with salary > 65000

```
SELECT id, first_name, last_name, department, salary, city
FROM employees
WHERE department = 'IT'
and city = 'Johannesburg'
and salary > 65000;
```

Output:

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Maidoo	IT	72000	Johannesburg